

Keystone Epitope Theory: Implications for the Investigation of Multiple Sclerosis

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Abstract

Multiple sclerosis (MS) has long presented a paradox: a strong and reproducible association with Epstein–Barr virus (EBV) exposure despite EBV’s near-universal prevalence. Large prospective cohorts show that EBV seroconversion precedes MS onset and is associated with a marked increase in risk ¹, while earlier work links risk to prior infectious mononucleosis and to elevated anti-EBV nuclear antigen 1 (EBNA1) antibody responses ^{2,3}. Yet EBV is clearly not sufficient, implying additional “gates” governed by host genetics, developmental timing, tissue context, and the quality of antiviral control.

Keystone Epitope Theory (KET) proposes that persistent, human-adapted pathogens such as EBV imprint high-gain immune priorities toward constrained peptide-HLA geometries (“keystone epitopes”), creating plausible vulnerabilities when modified self or foreign antigens accidentally mimic those geometries in the right HLA and niche ⁴⁻¹⁰. Applied to MS, KET reframes EBV association as a conditional recall problem and separates at least two related but nonidentical endpoints: liability to develop MS and quantitative control of the EBV reservoir. In this formulation, HLA-DRB1*15:01 may act mainly through qualitative class II presentation, altered-self presentation, and/or cross-reactive CD4 programs, whereas HLA-A*02:01 is more consistent with protective class I antiviral surveillance. Other class I backgrounds, including HLA-A*03:01 and HLA-B*07, may define relatively higher-risk or impaired-control contexts, although direct CNS cross-reactivity of the relevant class I clonotypes remains unproven. Blood EBV DNA, antibody features, exhaustion, and CNS-residency phenotypes measured after onset may be state-dependent consequences or modifiers and should not automatically be interpreted as the causal predisposition.

The review summarizes key evidence for EBV-focused immunity in MS, including host–virus genetic convergence ¹¹, and integrates recent findings supporting several nonexclusive routes from EBV to CNS autoimmunity: direct EBNA1–CNS mimicry, EBV-induced remodeling of HLA-

DR15 self-peptide presentation, survival signals that sustain antigen-capturing B cells within CNS niches, and alterations in antiviral cellular and humoral responses^{12-17,18,19}. It then proposes a lesion-anchored, receptor-first workflow that defines the cognate peptide–HLA surface before testing structural mimicry, altered self-presentation, niche-dependent amplification, and post-translationally modified targets, treating the CNS niche as the gate where specificity, antigen pressure, and inhibitory tone converge. The endpoint-separated model also has broader implications for autoimmunity by distinguishing class II-dependent immune focusing and humoral organization from class I-dependent antigen-source control or tissue execution, while treating tissue access and post-onset immune state as separate endpoints. The model also treats focal blood-brain barrier (BBB) activation as a distinct local access gate: gadolinium enhancement can precede conventional lesion evolution, but the temporal sequence linking barrier opening to entry of latently EBV-infected memory B cells remains unproven.

Keywords: multiple sclerosis; Epstein-Barr virus; Keystone Epitope Theory; autoimmunity; HLA class I and II; MHC-I-opathy; EBV reservoir control; molecular mimicry; altered-self presentation; T-cell receptor repertoire; intrathecal immunity; immunoepitidomics; blood-brain barrier

Key messages

- The epidemiological evidence strongly supports EBV exposure as a near-necessary antecedent in most MS cases studied to date, but the mechanism remains the central gap^{1,20}.
- Recent studies support multiple nonexclusive EBV-linked pathways, including direct EBNA1–CNS T-cell cross-reactivity, EBV-induced HLA-DR15 presentation of myelin peptides, B-cell-dependent antigen persistence and local restimulation, and altered class I, late-lytic CD4, and antibody Fc responses¹²⁻¹⁹.
- KET compresses the mechanistic search space by prioritizing the “time and site of the crime”: expanded lesion/CSF TCR clonotypes and their restricting peptide–HLA surfaces.
- A practical KET workflow is receptor-first and endpoint-aware: identify the restricting HLA and cognate peptide–MHC surface first, distinguish susceptibility mechanisms from EBV-reservoir-control mechanisms, and then test direct mimicry, EBV-induced altered self-presentation, and protective versus pathogenic antiviral surveillance^{17-19,21,22}.
- MS provides a tractable model for testing a two-HLA-axis architecture in which class II organizes upstream CD4–B-cell specificity and humoral responses, while class I modifies control of the persistent antigen source, direct tissue recognition, or both. The same architecture should be tested, rather than assumed, in other autoimmune diseases.
- Focal BBB permeability is an early feature of many newly forming lesions and can precede conventional MRI abnormalities or symptoms, but neither a generalized preclinical barrier defect nor a fixed sequence of BBB opening followed by entry of latently EBV-infected memory B cells has been established²³⁻²⁵.

1. Problem statement: the unmet mechanistic need

MS risk is shaped by a familiar triad: genetics (particularly HLA class II), environment (vitamin D and UV exposure, smoking), and infection history. EBV sits at the intersection. However, the field still lacks a tractable, testable map that links EBV immunity to the defining pathology of MS: compartmentalized CNS inflammation, blood–brain barrier perturbation, demyelination, and progressive neurodegeneration.

KET's claim is not that EBV is simply a trigger, but that EBV acts as a calibrator of immunodominance hierarchies. In this view, autoimmunity can reflect a failure of prioritization under constraints rather than an indiscriminate collapse of tolerance⁶⁻⁸. The practical question becomes: which EBV-trained clonotypes, recognizing which peptide–HLA geometries, are executing injury in the CNS niche? Figure 1 summarizes the overall KET-MS logic, including the four-condition gate and the post-initiation consequences.

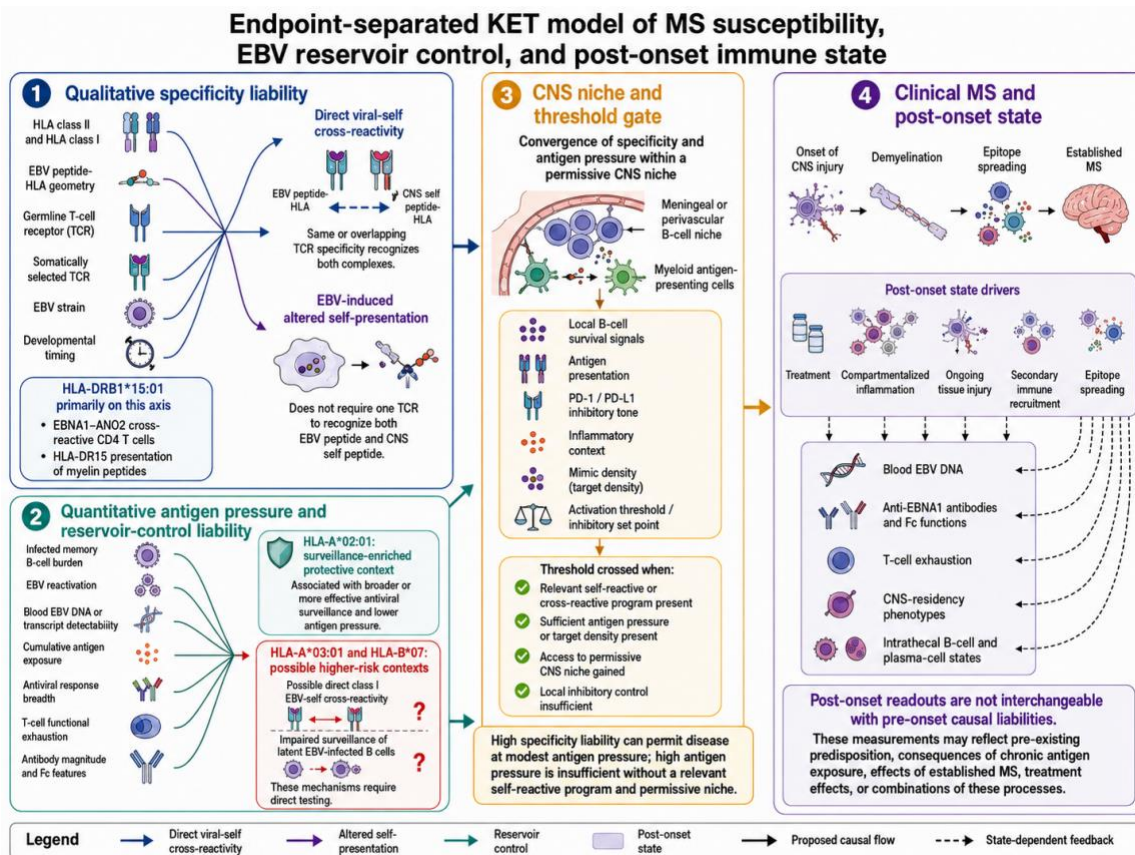


Figure 1. Endpoint-separated KET model of MS susceptibility, EBV reservoir control, and post-onset immune state. A qualitative specificity liability determines whether EBV-trained immunity can engage CNS self through direct molecular mimicry or EBV-induced altered-self presentation. A partially independent antigen-pressure liability determines the quantity and duration of EBV antigen exposure through the infected B-cell reservoir, antigen processing, HLA class I and II, and antiviral response quality. These liabilities become clinically consequential only in a permissive CNS niche in which local antigen presentation and inflammatory context exceed inhibitory control. Measurements made after MS onset, including blood EBV DNAemia, anti-EBNA1 features, exhaustion, CNS residency, compartmentalization, treatment effects, and epitope spreading, may be consequences or modifiers rather than the original causal liability^{26 27 17-19,22}. The CNS-niche gate includes endothelial activation and focal BBB access, which can precede visible lesion evolution but does not identify the initiating leukocyte population or establish that EBV-infected B cells are the first entrants^{23 24}.

With that map in place, the next sections lay out the empirical constraints and suggest how they can be translated into a testable discovery workflow.

What / Why / How (KET lens)

Table 1. What / Why / How framing of Keystone Epitope Theory applied to multiple sclerosis.

Question	KET interpretation	Implication for MS investigation
What is the problem?	We can describe EBV association and HLA risk, but cannot yet specify the initiating peptide–HLA geometry or distinguish mechanisms that mainly affect MS susceptibility from those that mainly affect quantitative control of the EBV reservoir.	Rather than searching broadly for ‘autoantigens’, start by mapping lesion or CSF effector receptors to their cognate pMHC and then determine whether the relevant host-virus context aligns with susceptibility, reservoir control, or both.
Why does it arise (development + evolution)?	EBV is a long co-adapted persistent virus. Early-life imprinting tends to occur in a tolerizing milieu; delayed, inflammatory primary infection may distort clonotype selection and regulatory set points.	Treat timing of EBV acquisition (IM vs early asymptomatic infection) as a modifier of the “activation threshold” gate.
How do we fix the knowledge gap?	Identify the keystone epitope first using TCR-forward mapping and restricting HLA, then search for CNS mimics and PTMs that recreate that surface at sufficient density.	A bounded, experimentally falsifiable workflow that can converge on a seed epitope before epitope spreading obscures the initiating event, while separating qualitative mimicry effects from quantitative reservoir-control effects.

2. Empirical constraints: what the data force us to explain

2.1 EBV exposure and MS risk precede disease onset

Prospective cohorts demonstrate that EBV seroconversion precedes MS onset and is associated with a large increase in MS risk, supporting temporality and specificity ¹. Earlier prospective studies have shown that elevated anti-EBV antibody titers (including EBNA responses) can precede MS diagnosis ², and analyses of US military serum repositories support anti-EBV antibodies as informative serologic markers ²⁸.

Infectious mononucleosis (symptomatic delayed primary EBV infection) is associated with an approximately twofold increase in MS risk in meta-analyses ^{3,29}, and the excess risk persists over decades in large population datasets ³⁰.

Several recent studies suggest a mechanistic bridge between delayed primary EBV infection and subsequent formation of intrathecal immune niches relevant to MS pathogenesis. Symptomatic primary infection has been associated with B-cell subsets that acquire CNS-homing behavior and recruit inflammatory T cells, providing a plausible route from delayed infection to establishment of an intrathecal antigen-amplifier compartment ³¹. Consistent with that interpretation, EBV persistence in B cells has been associated with expansion and functional bias of CXCR3⁺ memory B-cell subsets with brain-homing features and enhanced anti-EBNA1 plasma-cell differentiation in MS contexts ³².

Complementing these trafficking studies, analysis of deep cervical lymph nodes from newly diagnosed, untreated individuals with MS identified increased memory B-cell subsets, double-negative memory B cells with a lytic-EBV-like transcriptional state, EBV-targeting memory CD8 T cells in a subset, and increased EBV DNA in cervical nodes and saliva ¹². These cross-sectional data do not establish causality, but identify a peripheral draining compartment in which EBV activity, B-cell state, and antiviral T-cell responses can be examined together.

A 2026 population-based report described individuals whose recorded MS onset preceded laboratory evidence interpreted as primary EBV infection ³³. The report challenges an absolute formulation in which documented EBV infection is required before every case of MS, although retrospective serologic classification may not fully distinguish remote infection, primary infection, and reactivation. Independent replication and antigen-resolved longitudinal sampling will be needed to determine whether these cases represent an EBV-independent minority endotype, misclassified prior infection, or an alternative disease pathway.

Adenotonsillectomy was also associated with increased odds of pediatric-onset MS and a higher subsequent relapse rate in an observational study ³⁴. This finding is compatible with a role for tonsillar and adenoidal lymphoid niches in EBV acquisition, immune imprinting, or reservoir control, but confounding by indication and the nonrandomized design preclude causal interpretation.

2.2 EBNA1 immunodominance and risk stratification with HLA

Multiple datasets implicate EBNA1 as a central antigen in the EBV–MS relationship. Elevated EBNA1 IgG titers are associated with increased MS risk, and antibody reactivity to specific EBNA1 regions has shown particularly strong associations ³⁵. In a population-based case-control cohort, combining HLA-DR risk profiles with EBV serology substantially improved discrimination between MS and controls ³⁶.

Although HLA-DRB1*15:01 remains the dominant HLA association, high-resolution analyses support independent class I effects, including protection associated with HLA-A*02:01-containing haplotypes ²⁶. These signals should be interpreted against distinct endpoints: liability to develop MS, quantitative control of the circulating EBV reservoir, and immune phenotypes measured after MS onset are related but are not interchangeable.

Two population-scale sequencing studies published in 2026 sharpen this distinction. Schmidt and colleagues identified shared HLA class I genetic architecture between detectable blood EBV reads and MS that was driven largely by HLA-A*02:01; importantly, HLA-A*02:01 was associated with better EBV control as well as lower MS risk, whereas no consistent effect of HLA-DRB1*15:01 on EBV-read positivity was observed ²². Nyeo and colleagues independently implicated antigen-processing pathways and both class I and class II peptide-presentation variation in persistent blood EBV DNA; HLA-A*03:01 was associated with increased DNAemia, while stronger predicted class II presentation was associated with lower DNAemia ²⁷. These blood whole-genome-sequencing measures are state-dependent surrogates of detectable circulating EBV DNA, not direct measurements of total-body latent reservoir, lymph-node or CNS antigen exposure, or the initiating pathogenic event in MS.

Together, these findings support an endpoint-specific HLA gate. HLA-DRB1*15:01 may act predominantly through qualitative class II biology, including response focusing, direct cross-reactive CD4 programs, or EBV-induced altered-self presentation. HLA-A*02:01 is more consistent with broad and effective antiviral surveillance that reduces cumulative antigen pressure. HLA-A*03:01 and HLA-B*07 may represent higher-risk or relatively impaired-control contexts, but any direct CNS cross-reactivity must be demonstrated rather than inferred from association. Consistent with this distinction, HLA-B*07-restricted EBV-specific CD8 T cells were reported to recognize a narrower EBV epitope set and to exhibit more exhausted and CNS-residency-associated phenotypes than HLA-A*02-restricted cells; anti-EBNA1 IgG was highest in HLA-B*07-positive, HLA-A*02-negative MS ¹⁷. These case-associated phenotypes may reflect predisposing antiviral architecture, chronic antigen exposure after disease initiation, or both.

Anti-EBNA1 IgG should likewise not be treated as a simple viral-load measurement. It is a composite readout influenced by cumulative antigen exposure, class II presentation and CD4 help, B-cell state, and antibody selection. The same serologic phenotype could therefore arise through different combinations of specificity liability and antigen-pressure liability.

A distinct preclinical humoral endotype further supports separation of disease predisposition from later tissue injury. Approximately 10% of individuals who later developed MS carried a stable autoantibody signature against a shared peptide motif years before symptom onset, accompanied by higher serum neurofilament light concentrations³⁷. Subsequent assay development identified an MS-derived monoclonal antibody to this motif and implicated the EBV protein BRRF2 and the human intermediate-filament protein vimentin among candidate tolerance-breaking and cross-reactive antigens³⁸. This is unlikely to define all MS, but it provides a tractable example of an antigen-resolved preclinical endotype.

Longitudinal serum proteomics provides a complementary temporal sequence. Evidence of myelin injury was detectable approximately seven years before symptomatic MS, preceding evidence of axonal injury by about one year, whereas astrocyte-associated changes emerged closer to clinical onset³⁹. Individuals with the predictive autoantibody signature also showed greater preclinical immune activity. These observations support a multistage model in which antigen-specific humoral or T-cell dysregulation can precede measurable tissue injury, which in turn precedes overt neurological disease.

2.3 Environment and other modifiers act like ‘threshold knobs’

Vitamin D, UV exposure and smoking are reproducible MS modifiers, and plausibly intersect with EBV immune control. A vitamin D response element in the HLA-DRB1*15:01 promoter suggests a direct mechanistic link between vitamin D biology and the major genetic risk locus⁴⁰. Smoking provides an example of context-dependent risk modification within a gated model (Figure 6): it interacts with MS HLA risk architecture to amplify risk⁴¹. Smoking has also been associated with higher anti-EBNA1 titers in MS-relevant epidemiologic analyses, aligning with the idea that peripheral inflammatory priming can intensify EBV-linked immune programs⁴². Genomic studies indicate that EBV transcriptional regulators, particularly EBNA2, occupy autoimmune susceptibility intervals including those associated with MS and intersect with host regulatory architecture such as vitamin D receptor occupancy, creating a plausible surface for gene-environment-virus interactions that modulate threshold control rather than acting as single-cause triggers^{43,44}. Beyond HLA, large-scale MS genetic mapping implicates regulatory programs across peripheral immune cells and CNS-resident microglia, consistent with a model in which breach amplification and downstream injury involve both infiltrating lymphocytes and CNS-resident antigen-presenting circuits once the gate is crossed⁴⁵.

Consistent with a developmental and threshold-based model, analyses in two case-control cohorts found that EBV indices, lower vitamin D, low sun exposure, and HLA-DRB1*15:01 converged on shared DNA-methylation modules associated with MS onset, with mediation estimates that were directionally consistent across cohorts⁴⁶. These data do not prove that the methylation states are causal, but they provide a plausible molecular layer through which infection history, environment, and HLA may jointly influence immune calibration.

A further, still under-tested axis is germline diversity in antigen receptors themselves, particularly TCR loci, and how this diversity may interact with EBV strain variation across human populations.

2.4 Germline TCR variation, EBV strain diversity, and discordant coevolution: an underused axis for gating

KET treats HLA as a principal “hardware” constraint, but peptide-HLA recognition is a two-body problem: the other half is the receptor. The germline TCR loci (TRA/TRB/TRG/TRD) contain coding polymorphisms and haplotypic structure that can alter V-gene sequence features (including CDR1/CDR2 chemistry) and bias which peptide–HLA geometries are most easily generated, selected, and expanded ^{47,48}. This matters for a TCR-forward strategy in MS because “public” or stereotyped TCRs can be partly germline-enabled, not purely antigen-selected, and failure to account for germline TR alleles can distort V-gene assignment and cross-cohort comparisons.

Historically, several candidate-gene and linkage studies have suggested that polymorphisms near TCR loci may contribute to MS susceptibility. These included reported associations or linkage signals involving TCR β or TCR α/δ markers ^{49,50 51,52}. However, replication was inconsistent, and multiple cohorts reported no convincing association between MS and TCR gene polymorphisms across α , β , γ , and δ chains ⁵³⁻⁵⁵. A later synthesis emphasized that TR loci are structurally complex and historically poorly captured by standard marker sets, so the net message is best framed as “not established” rather than “excluded,” with the additional warning that population structure and technical ascertainment may have contributed to both positive and negative findings ⁵⁶.

However, the feasibility of such studies is now improved. High-throughput repertoire sequencing can infer germline TR alleles directly from AIRR-seq data and has revealed extensive TRB germline variability, including alleles absent from standard references and showing ancestry stratification ^{57,58}. Complementary approaches using improved genomic resources and targeted TR locus sequencing continue to expand the known catalog of germline TR alleles, revealing substantial previously unrecognized diversity that was poorly captured by earlier reference sets ⁵⁹. More broadly, given the well-established out-of-Africa gradient in human genetic diversity, it remains plausible that germline TR diversity is similarly structured across populations, although current TR-focused datasets do not yet distinguish between founder effects, selection, or coevolution. This broader framing is supported by population-genetic analyses showing a serial founder effect and monotonic loss of genetic diversity with distance from Africa across many human loci ^{60,61}.

Earlier population surveys also documented dense SNP variation across TCR loci and substantial allele frequency differences among African American, Chinese, Mexican and Northern European samples, with patterns broadly compatible with demography while not excluding selection or co-evolution ^{62,63}.

In parallel, EBV shows pronounced phylogeographic structure and deep type 1/type 2 divergence (largely defined by EBNA2/EBNA3) with linked variation extending into additional loci, including genes involved in B-cell tropism and immunogenic targets ⁶⁴⁻⁶⁶. Genome-wide analyses recover geographically stratified EBV subpopulations and enrichment of linked variation in immunogenic genes, consistent with a role for host immune pressure and local adaptation, while also making a key methodological point: it is difficult to disentangle host demography and sampling structure from true selection in a polymorphic virus ⁶⁷. Moreover, genome-to-genome analyses show that human genetic variation can correlate with EBV sequence variation, suggesting that host factors can shape EBV diversity at measurable scales⁶⁸. A related “discordant coevolution” concept has been explicitly proposed in other EBV-associated contexts in admixed populations ⁶⁹.

A recent pilot study also established the feasibility of recovering near-complete wild-type EBV genomes from individuals with MS. Near-complete genomes were obtained from 9 of 20 participants with MS and 4 of 20 controls, revealing extensive sequence diversity; in silico analyses suggested that variants within an EBNA1 cross-reactive region could alter predicted HLA binding ⁷⁰. The small sample and ex vivo enrichment method preclude conclusions about disease association, but the study provides a practical basis for larger strain-resolved analyses.

Taken together, there are at least three plausible explanations for correlated geography in HLA/KIR, germline TR alleles, and EBV strain variation: founder effects during migrations, simple selection operating on each component without coadaptation, or genuine coevolution and “matched” equilibrium states between host receptors and locally circulating EBV lineages. At present, these explanations cannot be cleanly distinguished, so strong claims should be avoided. However, the general principle has precedent. In *Helicobacter pylori*, discordant pairing of host ancestry and bacterial ancestry associates with increased gastric inflammation and pre-cancerous lesions, whereas matched host–pathogen ancestry is comparatively benign, implying that disruption of coevolved relationships can modulate disease risk ⁷¹.

For MS, KET turns this into a testable modifier of the four-condition gate: not only whether EBV was acquired, but whether the infecting EBV strain’s keystone-relevant epitope geometries align with the host’s HLA and germline TCR landscape in a manner compatible with regulated equilibrium. TCR-forward mapping of expanded lesion or CSF clonotypes should therefore be paired with germline TR allele calling, high-resolution HLA typing, and, where feasible, EBV strain sequencing. A null result would still be informative; a reproducible host–virus interaction would provide a mechanistically informative axis of discordant coevolution for integration into the gated model.

3. EBV biology relevant to a keystone model

EBV persists for life in the B-cell compartment with tightly regulated latency programs. The infected memory B-cell pool provides long-term persistence, with periodic transitions through proliferative programs and reactivation. EBNA1 is required for maintenance and segregation of

the EBV episome in dividing cells, and its immune visibility is shaped by immune evasion mechanisms that can bias presentation toward class II pathways ²⁰. The relevant latency programs and where EBNA1 persists across B-cell differentiation are summarized in Figure 2.

Recent antigen-resolved data broaden the EBV program relevant to MS beyond latent antigens. In untreated MS, CD4 T-cell responses were approximately twofold greater than in controls and were directed predominantly toward late lytic capsid and glycoprotein antigens; these responses and salivary EBV shedding declined after anti-CD20 therapy ¹⁴. These observations link B-cell depletion to reduced lytic antigen exposure, but do not establish whether the responding CD4 T cells are pathogenic, protective, or markers of ongoing viral activity.

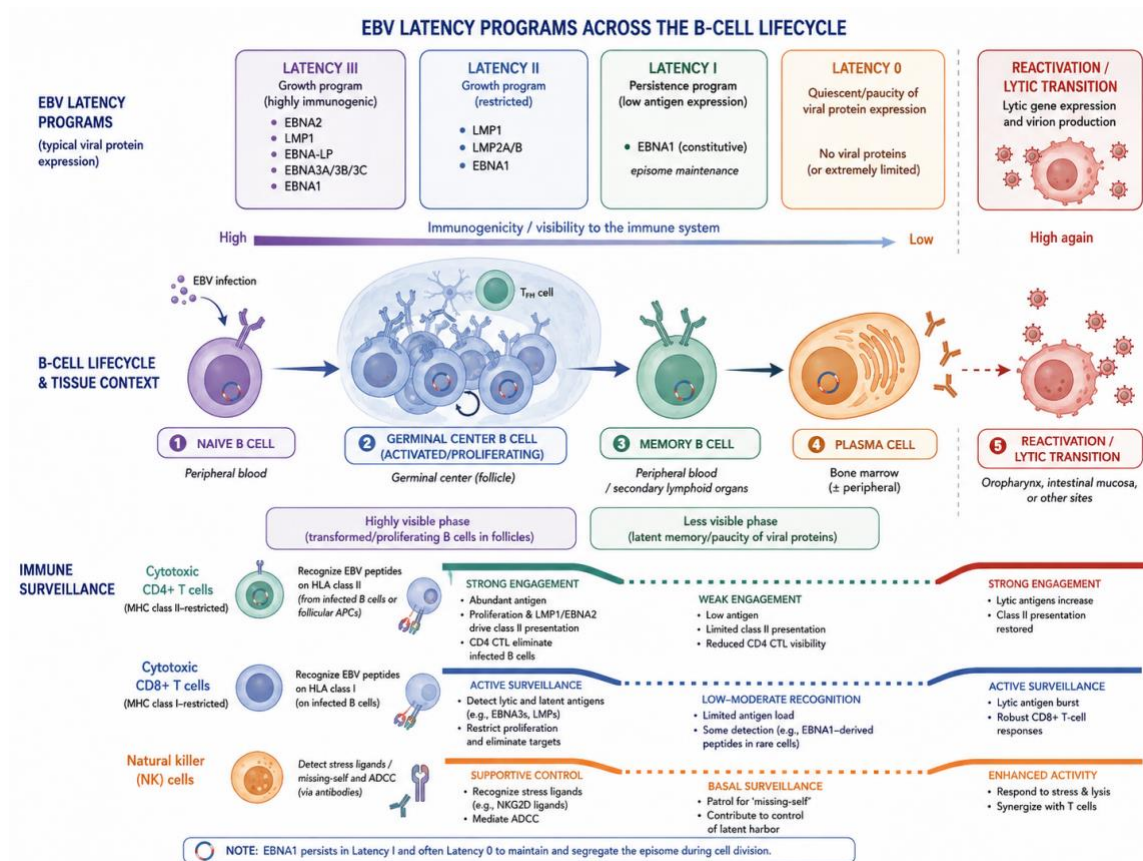


Figure 2. EBV latency and reactivation across B-cell differentiation. Simplified schematic of dynamic EBV programs across B-cell states: latency III in activated or proliferating B cells; latency II during germinal-center transit; latency 0 in resting memory B cells; transient latency I during memory-B-cell division; and lytic reactivation during plasma-cell differentiation. Class II and class I presentation can engage EBV-specific CD4 and CD8 T cells, respectively. Recent findings of enriched late-lytic antigen-specific CD4 responses in MS emphasize that both latent and lytic programs should be interrogated ^{14,20}.

In the KET framework, EBV provides a chronic, anatomically organized antigen source within follicular niches, making both helper and cytotoxic control biologically relevant. For MS, a

central unresolved question is whether follicular EBV control is mediated predominantly by EBV-specific cytotoxic CD4 T cells, by CD8 T cells, or by a coordinated program involving both, and whether these effector pathways can be redirected by peptide mimicry. This leads to the related question of how B-cell-centered persistence intersects with the CNS compartment.

4. CNS access and compartmentalization: the blood-brain barrier, B cells, follicles, and oligoclonal IgG

4.1 Focal blood-brain barrier activation as an early lesion gateway

Because gadolinium contrast normally remains intravascular, enhancement reports focal BBB permeability. Serial gadolinium-enhanced MRI studies have held up in showing that this permeability is an early feature of many newly forming inflammatory MS lesions. In the seminal serial study, contrast enhancement preceded other MRI abnormalities in four cases and preceded clinical evidence of the new lesion in one ²³. Dynamic contrast-enhanced MRI subsequently showed that early lesions can begin with centrifugal leakage around a central vein and evolve over several days toward centripetal enhancement at the expanding lesion edge ²⁴. These observations establish BBB opening as an early phase of acute focal lesion development; they do not demonstrate that a generalized BBB defect universally precedes the clinical onset of MS.

BBB failure is better understood as activation and dysfunction of the neurovascular unit than as a simple physical rupture. Inflammatory cytokines and chemokines, endothelial NF- κ B signaling, oxidative stress, matrix metalloproteinases, adhesion-molecule expression, and remodeling of tight and adherens junctions can jointly increase permeability and leukocyte trafficking ^{25,72}. Human MS tissue shows abnormalities of endothelial tight-junction proteins in active lesions and in portions of normal-appearing white matter ⁷³, while patient-derived induced pluripotent stem cell models support an intrinsic component of brain endothelial dysfunction that can increase immune-cell migration ⁷⁴. The relative contributions of pre-existing endothelial susceptibility and inflammation-induced barrier dysfunction are likely to vary among lesions and disease stages.

Gadolinium enhancement does not identify which leukocyte population initiates the local process or establish a fixed order of cellular entry. Contrast leakage and leukocyte diapedesis are related but not interchangeable endothelial functions; changes in adhesion and paracellular or transcellular migration need not scale directly with the gadolinium signal. Activated T cells and monocytes may be early participants in endothelial activation and transendothelial migration, while B cells may enter during or after the gateway becomes permissive. To our knowledge, no lesion-by-lesion longitudinal human study has demonstrated that BBB opening is followed specifically by entry of latently EBV-infected memory B cells. The BBB should therefore be treated as a dynamic access and amplification gate rather than as a downstream event that must be initiated by EBV-positive B cells ²⁵.

Latently infected or otherwise EBV-affected memory B cells nevertheless remain plausible entrants and local antigen-presenting cells once CNS access is available, particularly given evidence that symptomatic primary EBV infection can induce CNS-homing B-cell phenotypes³¹. EBV can also infect cultured human brain microvascular endothelial cells and induce proinflammatory mediators and leukocyte adherence⁷⁵, but this experimental observation does not establish endothelial EBV infection as a causal event in human MS. In the endpoint-separated KET model, focal BBB activation is therefore part of the CNS niche and threshold gate: it may arise through T-cell, innate, endothelial, or systemic inflammatory mechanisms and then permit EBV-affected B cells, cross-reactive T cells, or both to sustain compartmentalized inflammation.

4.2 B-cell niches, local restimulation, and oligoclonal IgG

CSF oligoclonal bands (OCBs) are present in the majority of MS cases and reflect clonal expansion of intrathecal B cells and plasmablasts^{76,77}. Meningeal lymphoid aggregates and follicle-like structures have been described in a subset of MS cases and provide a plausible intrathecal ‘antigen-amplifier’ niche. In secondary progressive MS, their presence has been linked to more severe cortical pathology and earlier disease onset, supporting the niche-seeding logic that the gated KET model later operationalizes⁷⁸. Two issues are often conflated: (i) whether meningeal follicle-like structures occur and associate with cortical injury, and (ii) whether EBV is consistently detectable within those structures and adjacent CNS tissue. A widely cited positive neuropathology study reported EBV-infected B cells and plasma cells in MS brains and identified meningeal follicles as major sites of EBV persistence⁷⁹. However, the extent to which EBV-infected B cells are consistently present in CNS tissue remains contested across laboratories^{80,81}.

Current evidence favors a role for meningeal ectopic lymphoid structures in restimulating and sustaining previously primed memory or effector T-cell responses rather than serving primarily as sites of de novo naive T-cell priming. In experimental CNS autoimmunity, cognate T–B-cell interactions within ectopic follicles sustained proinflammatory T-cell states and B-cell differentiation⁸², while human MS studies have shown that memory B cells can drive HLA-DR-dependent proliferation of brain-homing autoreactive CD4 T cells⁸³. These findings refine the “antigen-amplifier” concept: the meningeal niche is most plausibly a compartment for local recall, survival, and epitope spreading. In KET terms, it functions as the third gate, where receptor specificity, antigen density, and local survival or checkpoint signals converge. Neuropathological analysis of perivascular cuffs in advanced MS likewise associated greater B-cell abundance with active or mixed active/inactive lesions and with local CD4 T-cell and antibody-secreting-cell interactions. LMP1-positive B cells were sparse within cuffs, indicating that local inflammatory amplification can occur without abundant detectable EBV-positive cells at every active site¹⁵.

Two recent studies provide complementary mechanisms by which EBV-affected B cells could amplify CNS autoimmunity. EBV infection reprogrammed the B-cell transcriptome and the HLA-

DR15 immunopeptidome, increasing presentation of myelin basic protein-derived peptides and activation of myelin-reactive CD4 T cells ¹⁹. In a mouse model, CNS-infiltrating myelin-reactive B cells captured antigen from neural parenchyma and were normally short-lived; CD40L signaling or expression of the EBV latent membrane protein LMP1 bypassed this checkpoint, permitting local plasma-cell differentiation and demyelinating lesion formation ¹⁶. The latter findings are model-based rather than direct evidence of LMP1-driven human MS, but they identify a testable niche-specific survival and amplification pathway.

Claims that OCBs are predominantly EBV-specific have been reported ^{84,85}, but other studies support a polyspecific intrathecal antibody response and caution against assuming EBV specificity of OCBs as a general rule ⁸⁶. The KET implication is straightforward: antibody specificity is informative, but the decisive step for tissue injury is more likely the effector TCR recognition event at the CNS site. High-dimensional antibody profiling further found that EBNA1-specific IgG in MS had enhanced Fcγ-receptor binding and proinflammatory effector functions, while Fcγ-receptor binding by EBNA1- and gp350-specific antibodies tracked clinical relapse or contrast-enhancing lesion activity ¹³. These associations do not prove antibody-mediated injury, but suggest that functional Fc properties may be more informative than antibody titer alone.

5. The effector arm: EBV-specific T cells and the special case for cytotoxic CD4

Multiple studies show that MS CSF contains expanded, patient-private T-cell clones, and that T cells in CSF can recognize autologous EBV-transformed B cells ⁸⁷. Primary EBV infection can elicit oligoclonal expansions of EBV-specific CD4 T cells with cytotoxic effector programs detectable at the single-cell level, consistent with a genuine class II-restricted CD4 cytotoxic arm rather than CD4 “help” alone ⁸⁸. EBNA1-reactive cytolytic CD4 clones have also been shown to inhibit proliferation of newly EBV-infected B cells via HLA-DR-restricted recognition early after infection ⁸⁹. These observations provide direct biological support for the KET premise that a cytotoxic, EBV-trained class II-restricted program exists and could become pathogenic if a CNS peptide–HLA surface functionally mimics a keystone EBV geometry in an inflammatory context. Current data leave open, but do not yet prove, that some HLA-DR15-restricted cytotoxic CD4 clonotypes may directly cross-recognize neuronal or glial ligands in particular MS endotypes.

An optimized antigen-resolved assay published in 2026 found that CD4 T cells from untreated individuals with MS preferentially recognized late lytic viral-particle antigens rather than latent antigens, whereas EBNA1 primarily activated CD8 T cells in that assay ¹⁴. This result broadens the candidate antigenic program beyond EBNA1 and reinforces the need to classify EBV-reactive clonotypes by antigen, HLA restriction, tissue localization, and function rather than by EBV specificity alone.

Recent work in paired blood/CSF analyses has reported that some expanded CSF T-cell clones are specific for autologous EBV-infected B cells ⁹⁰.

B-cell antigen presentation has distinct class II and class I consequences. MHC class II presentation can restimulate cognate CD4 T cells and amplify helper or cytotoxic CD4 programs, whereas MHC class I presentation of endogenous viral peptides renders EBV-infected B cells targets for CD8 T-cell surveillance ^{22,83}. The same B-cell compartment may therefore amplify class II-restricted recall while being constrained by class I-restricted antiviral control.

A well-established parallel literature shows that active MS lesions often contain dominant clonal expansions of CD8 T cells ⁹¹, and intrathecal repertoire studies have reported enrichment of EBV-reactive CD8 TCR sequences in MS CSF ⁹². This does not weaken the KET workflow; it strengthens the operational point. The “T cells at the time and site of the crime” may be CD8 and/or cytotoxic CD4 T cells, so receptor-first mapping should start agnostically from expanded lesion or CSF clonotypes and only then move upstream to keystone-epitope identification and CNS mimic discovery. HLA-resolved analysis further suggests that HLA-B*07-associated EBV-specific CD8 responses may be narrower, more exhausted, and more CNS-homing than HLA-A*02-associated responses ¹⁷. That pattern is compatible with impaired reservoir control or qualitatively different effector programming, but it does not by itself prove pathogenic CNS cross-recognition.

Antigen-resolved analysis of clonally enriched CSF CD8 T cells provides an important constraint on interpretation. Several expanded clonotypes recognized EBV antigens, including EBNA3A and BZLF1, but the tested EBV-specific TCRs did not recognize the screened homologous self-peptides ²¹. These clones may therefore represent antiviral surveillance of EBV activity within the CNS rather than direct molecular mimicry, although incomplete self-peptide coverage does not exclude cross-reactivity. EBV specificity alone should not be treated as evidence of pathogenicity.

KET emphasizes that these cytotoxic CD4 programs are not inherently autoreactive. The risk emerges when the same TCR geometries are engaged by CNS-presented mimics in the correct HLA context, and when local inhibitory set points (PD-1/PD-L1, CTLA-4, regulatory networks) are exceeded ⁶⁻⁸. Notably, EBV-infected B cells in MS brain samples have been reported to express PD-L1, suggesting a local immune-evasion microenvironment that could distort control and increase antigen burden ⁹³. A follicle-centric model for cytotoxic CD4⁺ control of EBV-infected B cells, and how the same program could be misdirected to CNS targets when the threshold is breached, is illustrated in Figure 3.

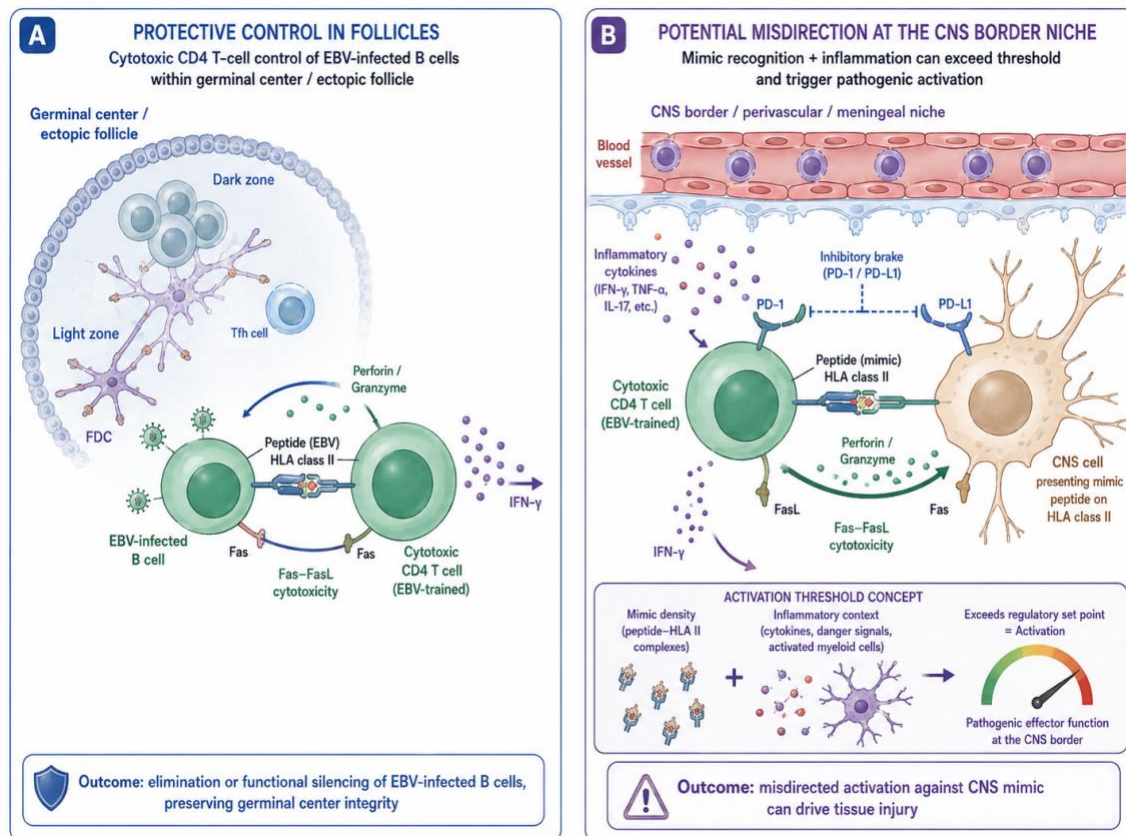


Figure 3. Follicular EBV control and potential cross-recognition at the CNS border. Panel A depicts class II-restricted cytotoxic CD4 T-cell recognition of an EBV-infected B cell within a follicular niche. Panel B depicts the testable possibility that a CNS-border cell presents a candidate mimetic peptide-HLA surface to an EBV-trained CD4 receptor when antigen density and inflammatory context exceed local inhibitory control. Parallel HLA class I-restricted CD8 surveillance of infected B cells is addressed in the text but is not shown in this simplified placeholder. EBV specificity alone does not establish CNS cross-reactivity or pathogenicity.

As shown in Figure 3, the key open mechanistic question is what CNS-presented peptide-HLA surfaces could plausibly engage these EBV-trained receptors. In KET terms, molecular mimicry is the experimentally tractable bridge once the relevant TCR and restricting HLA are known.

6. Molecular mimicry: moving from EBV immunity to CNS targets

Among proposed mechanisms linking EBV immunity to CNS injury, receptor-anchored molecular mimicry currently has some of the strongest direct mechanistic support. A landmark study isolated CSF-derived monoclonal antibodies from MS and demonstrated molecular mimicry between EBNA1 and the CNS protein GlialCAM, supported by structural data and functional readouts⁹⁴. Follow-on work indicates that combined antibody reactivity to EBNA1 and GlialCAM-related mimics can differentiate MS patients from controls and may help define an EBNA1-mimicry-linked serologic endotype⁹⁵. Complementary Fc profiling suggests that EBV-specific antibody function, not only antigen binding, may associate with focal disease activity¹³.

Independent lines of evidence implicate EBNA1-related immunity to additional CNS antigens, including anoctamin-2 (ANO2), where immune reactivity associates with MS risk and interacts with HLA and EBNA1 antibody levels ⁹⁶. At minimum, these findings make EBNA1 a rational starting antigen for KET-guided receptor mapping. Figure 4 integrates EBV exposure, EBNA1-focused serology, HLA background, intrathecal oligoclonal IgG, and candidate CNS mimicry axes, emphasizing that direct receptor cross-reactivity, altered self-presentation, and compartmental amplification are not mutually exclusive.

More direct T-cell evidence now places ANO2 within a receptor-level mimicry pathway. ANO2-specific T-cell responses were frequent in MS, EBNA1 and ANO2 elicited cross-reactive CD4 T cells with overlapping receptor repertoires and activated or cytotoxic phenotypes, and ANO2-directed immunity worsened demyelinating disease in experimental models ¹⁸. These findings materially strengthen the EBNA1–ANO2 axis, while not establishing that it is the initiating mechanism in every individual with MS.

A recent synthesis emphasizes that EBNA1 contains a concentrated region of multiple candidate CNS mimicry axes, including GlialCAM, CRYAB, and ANO2, rather than pointing to a single universal autoantigen ⁹⁷. For KET, this multiplicity predicts endotypic heterogeneity: the relevant mimic should depend on the restricting HLA, receptor repertoire, EBV sequence, target-cell presentation pathway, and tissue niche.

Direct viral–self mimicry is not the only plausible route. EBV-driven remodeling of the HLA-DR15 immunopeptidome can increase presentation of self peptides without requiring sequence homology to a viral epitope ¹⁹. KET-guided experiments should therefore distinguish direct cross-recognition of viral and CNS peptide–HLA surfaces from EBV-induced amplification of an autoreactive self-peptide response.

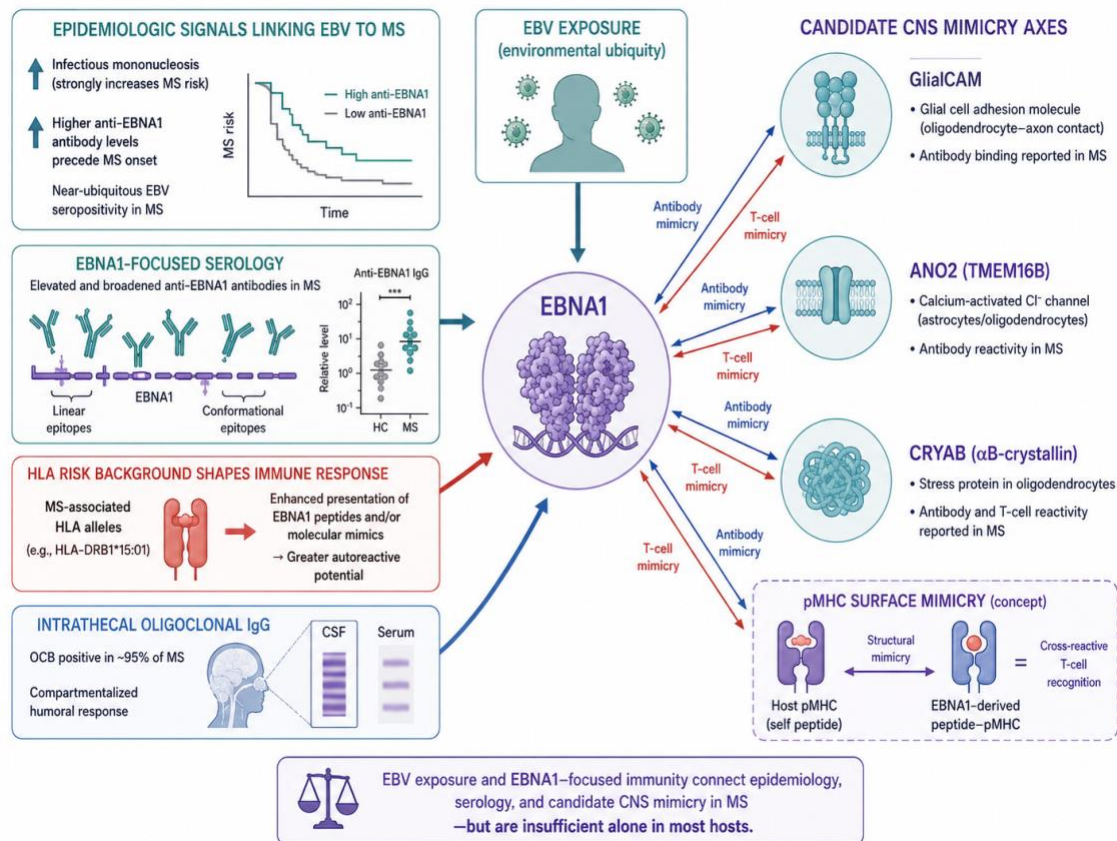


Figure 4. EBV exposure, EBNA1-focused immunity, and candidate CNS mimicry axes in multiple sclerosis. The figure integrates epidemiologic and serologic signals, HLA background, and intrathecal oligoclonal IgG with candidate EBNA1-linked CNS targets such as GlialCAM, ANO2, and CRYAB. Antibody mimicry and T-cell mimicry are shown as nonexclusive possibilities, and pMHC surface mimicry is illustrated as a structural concept. The figure is intended as a map of plausible endotypes rather than proof of a single universal initiating pathway^{13,18,94-96}.

These receptor-anchored mimicry datasets sharpen what may be necessary, but they still do not explain why most EBV-seropositive carriers of MS-associated HLA haplotypes never develop MS. KET addresses this gap by treating MS as a gated recall event that depends on niche seeding and a breach of local inhibitory thresholds, rather than on exposure alone. It also predicts that some post-onset EBV readouts, including viral DNA burden, exhaustion markers, antibody titers, and CNS-residency phenotypes, may reflect consequences or modulators of the established niche rather than the original causal liability.

7. KET applied to MS: a four-condition gate

KET is most useful when operationalized as a gated model. A workable MS version is a four-condition gate, analogous to conditional recall models in HLA-associated drug hypersensitivity. Importantly, the framework should separate at least two partially distinct outcome axes: liability to develop MS and efficiency of EBV reservoir control.

1. HLA constraint (the “hardware”): MS risk haplotypes constrain the peptide–HLA surfaces that can be stably presented and recognized. In current data, HLA-DRB1*15:01 most plausibly acts through qualitative class II presentation, altered-self presentation, and/or cross-reactive CD4 responses, whereas class I background, including the contrast between HLA-A*02:01 and higher-risk contexts such as HLA-A*03:01 or HLA-B*07, may more strongly shape antiviral surveillance and reservoir control ^{17,22,26,27}.
2. Keystone imprinting (the ‘training’): timing and inflammatory context of primary EBV infection shape clonotype selection, regulatory calibration, and tissue homing ³.
3. Niche access and seeding (the “where”): focal endothelial activation and BBB permeability can provide access to perivascular and meningeal compartments, after which EBV-affected B cells and previously primed T cells may be retained or restimulated. B-cell niches can sustain local recall and amplification even when EBV-positive cells are sparse at the sampled lesion ^{15,23,24,82,83}.
4. Threshold breach (the ‘why now’): sufficient mimic density, together with inflammatory context, overrides niche-specific inhibitory set points, enabling EBV-trained effector programs to attack CNS targets; once initiated, epitope spreading obscures the initiating seed. Survival and costimulatory signals that permit antigen-capturing B cells to persist may contribute to this breach ¹⁶.

Operationally, the gate must be realized in an anatomical niche. Serial MRI supports focal BBB opening as an early phase of many new lesions, but does not establish that EBV-infected B cells cause the opening or are the first cells to enter ^{23,24}. The BBB and meningeal or perivascular interfaces should therefore be modeled as dynamic access and amplification sites in which endothelial state, receptor specificity, antigen pressure, B-cell survival, and local presentation jointly determine whether the inhibitory threshold is crossed. EBV-infected cells may contribute to this process, but need not be abundant at every active lesion ¹⁵. The crucial question is not simply whether EBV is detectable locally, but whether the niche permits sustained display of the relevant peptide-HLA ligand and effector engagement.

Together, the four conditions provide a plausible explanation for partial penetrance: EBV exposure and HLA risk are common, but MS may require a permissive imprinting history, a CNS-relevant niche, and a threshold-breach event (Figure 5A–B). Once injury begins, compartmentalization and epitope spreading broaden the apparent antigenic repertoire and obscure the initiating seed (Figure 8A). This is why pre-MS liabilities should be distinguished conceptually from post-onset readouts such as EBV DNA burden, antibody features, exhaustion, and CNS residency.

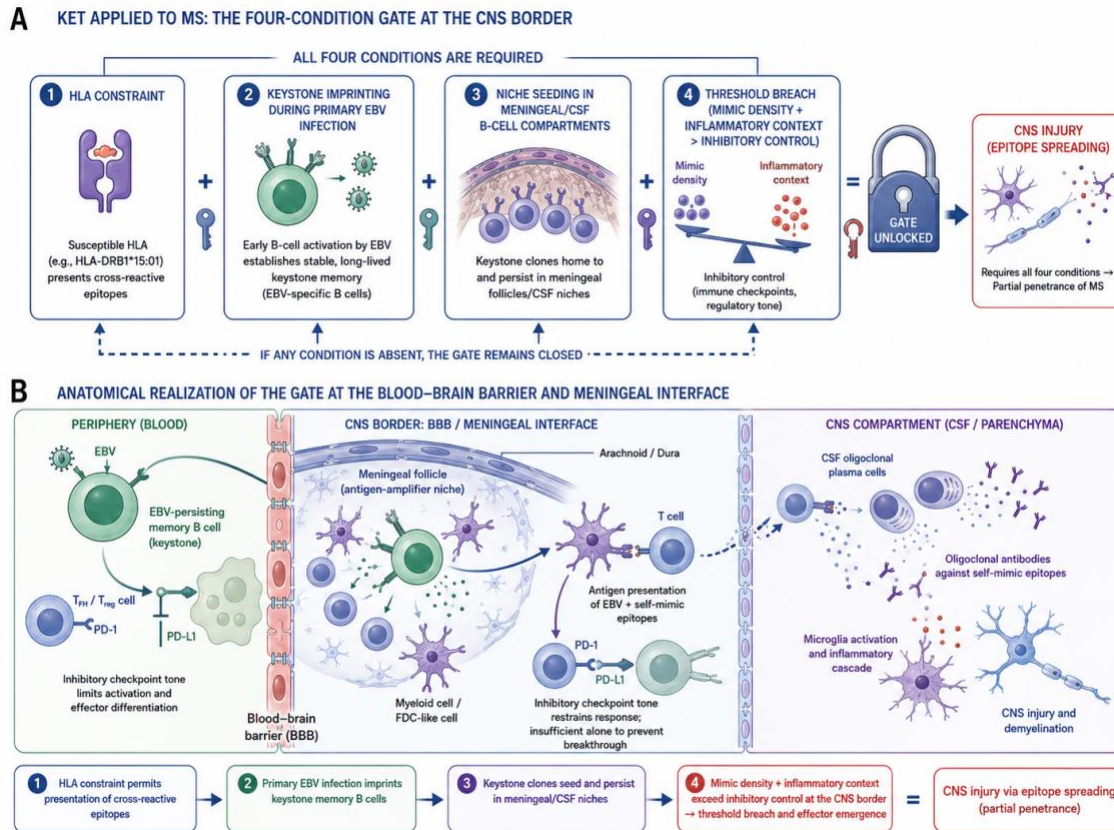


Figure 5. KET applied to MS as a four-condition gate at the CNS border. Panel A operationalizes KET as four interacting conditions: host receptor constraints; keystone imprinting; B-cell niche or reservoir access; and threshold breach. Panel B depicts a meningeal or perivascular niche in which B cells, CD4 and CD8 T cells, other antigen-presenting cells, and plasma cells can sustain local recall. The placeholder emphasizes anatomical realization of the gate; the endpoint-separated class I, class II, and post-onset distinctions are shown in Figure 1 and developed in Section 7. Such amplification may persist even when EBV-positive cells are sparse at the sampled lesion^{15,17,22,26,27,82,83}. Focal BBB activation is represented as a permissive access event within the third gate; the figure does not imply that EBV-positive B cells initiate barrier opening or are invariably the earliest cells to enter.

7.1 Candidate ‘threshold knobs’ and testable predictions

In KET terms, these factors function less like independent causes and more like tunable parameters that shift the probability of crossing a niche-specific inhibitory set point. Table 2 lists concrete, testable ‘threshold knob’ predictions; a simplified schematic is shown in Figure 6.

Table 2. Candidate ‘threshold knobs’ that modulate the probability of gate breach in KET-MS.

Modifier	KET interpretation	Testable prediction (examples)
Vitamin D/UV	May influence HLA-DRB1*15:01 expression and broader regulatory calibration.	Vitamin D status should be tested against HLA-DRB1*15:01 expression, regulatory phenotypes, and activation thresholds rather than assumed to act through a single pathway ⁴⁰ .
Smoking	Shifts inflammatory tone and may alter EBV antigen burden, antibody responses, or tissue susceptibility. HLA background modifies the observed risk association ^{98 41} .	Smoking, HLA class I and II, anti-EBNA1 features, and EBV activity should be modeled jointly as modifiers of penetrance ^{41,42,98} .
EBV strain, antigenic program, and persistent burden	Can alter epitope geometry, latent-versus-lytic antigen availability, and cumulative antigen pressure. Blood EBV DNA is a state-dependent surrogate of circulating reservoir control rather than a direct measure of the initiating CNS mechanism.	Wild-type EBV sequence, blood EBV DNA or transcript burden, and latent- or late-lytic response profiles should associate with distinct HLA- and TCR-defined immune states. Analyses should separate pre-onset susceptibility from post-onset disease activity ^{12,14,22,27,70,99,100} .
Host receptor architecture	HLA class I and II and germline TCR alleles constrain antiviral, self-reactive, and cross-reactive recognition, but different alleles may map to different endpoints. Class II risk may shape qualitative presentation and mimicry, whereas class I background may more strongly shape antiviral surveillance and reservoir control.	HLA-A*02:01 should, on average, associate with better EBV control and broader functional antiviral responses. HLA-A*03:01 and HLA-B*07 should be dissected into reservoir-control and specificity components; direct CNS cross-reactivity requires functional proof. Germline TR alleles should be included in cross-population analyses ^{17,22,26,27,57-68} .
Checkpoint tone (PD-1/PD-L1)	Controls local inhibitory set point within follicles and CNS borders.	High PD-L1 expression on EBV-infected B cells has been reported and may track with higher antigen burden and altered T-cell phenotypes in lesions ⁹³ .
Age / delayed primary infection	First imprint occurs in a more inflammatory milieu with different clonotype selection. Recent mechanistic work suggests that symptomatic primary EBV infection can generate CNS-homing B-cell subsets that recruit inflammatory T cells, providing a plausible 'Condition 3' bridge from	History of IM associates with broader EBNA1 epitope recognition and higher risk in specific HLA strata ^{3,36} .

Modifier	KET interpretation	Testable prediction (examples)
	infectious mononucleosis timing to intrathecal niche formation.	
BBB and endothelial state	Focal endothelial activation controls access to perivascular and meningeal niches and can be both an early lesion event and a consequence of local inflammation. It should be analyzed separately from EBV reservoir burden.	Dynamic imaging and endothelial assays should determine whether permeability precedes candidate T- or B-cell entry and whether EBV-affected B cells are necessary, permissive, or secondary. Gadolinium enhancement alone cannot identify the initiating cell population ^{23,24,74} .

A further, currently under-tested modifier is host-virus phylogeographic discordance combined with germline TCR variation, which could shift keystone epitope recognition and inhibitory set points without changing EBV serostatus (Section 2.4).

Persistent EBV DNA burden and its antigen-processing determinants may constitute an additional measurable modifier. Population-scale studies indicate that class I and class II effects differ by endpoint and cohort, so viral burden should be measured directly rather than inferred from an MS-risk HLA allele alone ^{22,27}.

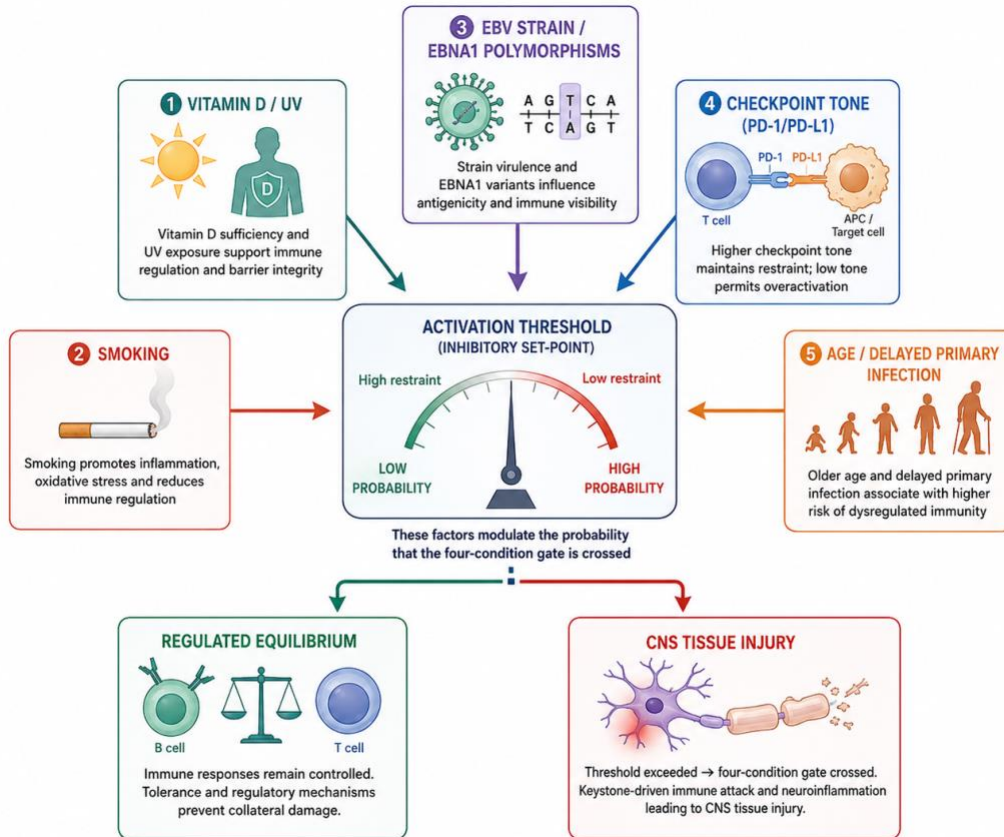


Figure 6. Modifiers of the probability of gate breach. Selected modifiers include age and timing of primary EBV infection; vitamin D, UV exposure, and smoking-related inflammatory tone; HLA class I and II context; EBV strain or antigenic program; and inhibitory checkpoint tone. These factors may influence either or both of the key axes in KET-MS: predisposition to disease initiation and quantitative control of the EBV reservoir.

7.2 Endpoint-separated KET-MS model: specificity, antigen pressure, and niche

The most parsimonious synthesis is a two-liability, one-niche model. One liability is qualitative: whether a particular EBV peptide-HLA-TCR or altered-self peptide-HLA-TCR geometry can engage CNS self. The second is quantitative: the cumulative amount and duration of EBV antigenic stimulation generated by imperfect control of infected B cells. Clinical MS is proposed to occur only when these liabilities converge in a CNS-accessible niche in which local antigen display and inflammatory context exceed inhibitory control. This is a conceptual heuristic rather than a fitted mathematical model.

Qualitative specificity liability. HLA-DRB1*15:01 may act mainly on this axis through preferential class II presentation, EBNA1-focused response quality, direct cross-reactive CD4 clonotypes, or EBV-induced presentation of self peptides such as myelin basic protein^{18,19,22}. Class I-restricted direct mimicry remains possible, including in HLA-A*03:01- or HLA-B*07-associated contexts, but has not been established and should require same-TCR, exact-HLA, and physiologic target-presentation evidence.

Quantitative antigen-pressure liability. HLA class I and II, antigen-processing genes, infected memory-B-cell burden, latent and lytic reactivation, and response breadth or exhaustion influence cumulative EBV antigen exposure ^{14,17,22,27}. HLA-A*02:01 appears to favor effective antiviral surveillance and lower detectable blood EBV DNA. HLA-A*03:01 and HLA-B*07 may contribute to less effective or narrower control in some settings. Elevated anti-EBNA1 IgG can arise from increased antigen pressure, class II immune focusing, or both, and is therefore not a direct substitute for viral-load measurement.

CNS niche and threshold liability. Neither cross-reactivity nor imperfect reservoir control is sufficient without an anatomical context that permits restimulation and effector function. Meningeal and perivascular B-cell-rich compartments can provide local antigen presentation, T-B cooperation, survival signals, and inflammatory amplification, while CD8 T cells may mediate much of the direct cytotoxic injury and CD4 T cells can provide help, inflammatory orchestration, or direct cytotoxicity ^{15,16,82,83}.

Pre-onset liability versus post-onset state. Once MS is established, blood EBV DNA, antibody titer and Fc function, T-cell exhaustion or CNS residency, B-cell compartmentalization, treatment, and epitope spreading are influenced by the disease state itself. They should therefore be treated as potential consequences or modifiers unless temporality is demonstrated. The model also predicts index-event selection among cases: individuals with a strong specificity liability may cross the disease threshold at relatively modest antigen pressure, whereas those with weaker specificity liability may require greater antigen pressure or a more permissive niche. This can produce weak or inverse correlations between the two liabilities within established MS despite both contributing to risk in the population.

Testable implications. HLA-DRB1*15:01 should predict antigen-specific quality more strongly than blood EBV DNA quantity. HLA-A*02:01 should associate with better reservoir control without serving as an absolute negative control for mimicry. HLA-A*03:01 and HLA-B*07 should be tested separately for impaired control and direct cross-reactivity. Lesion or CSF receptor-pHLA coherence should be more informative for tissue-destructive potential than blood DNAemia alone, and the highest risk should arise from cross-axis combinations of specificity liability, antigen pressure, and niche permissiveness.

8. Practical investigational workflow: lesion-anchored, TCR-forward mapping

The central methodological proposal is to invert the usual approach to autoantigen discovery. Rather than starting with a catalog of plausible CNS proteins and then hunting for responses, KET starts with the effector receptors at the site of injury and asks what they see. The resulting lesion-anchored, TCR-forward workflow is outlined in Figure 7. Because post-initiation epitope spreading rapidly reduces identifiability of the initiating seed, the approach prioritizes early sampling and receptor-first mapping (Figure 8A).

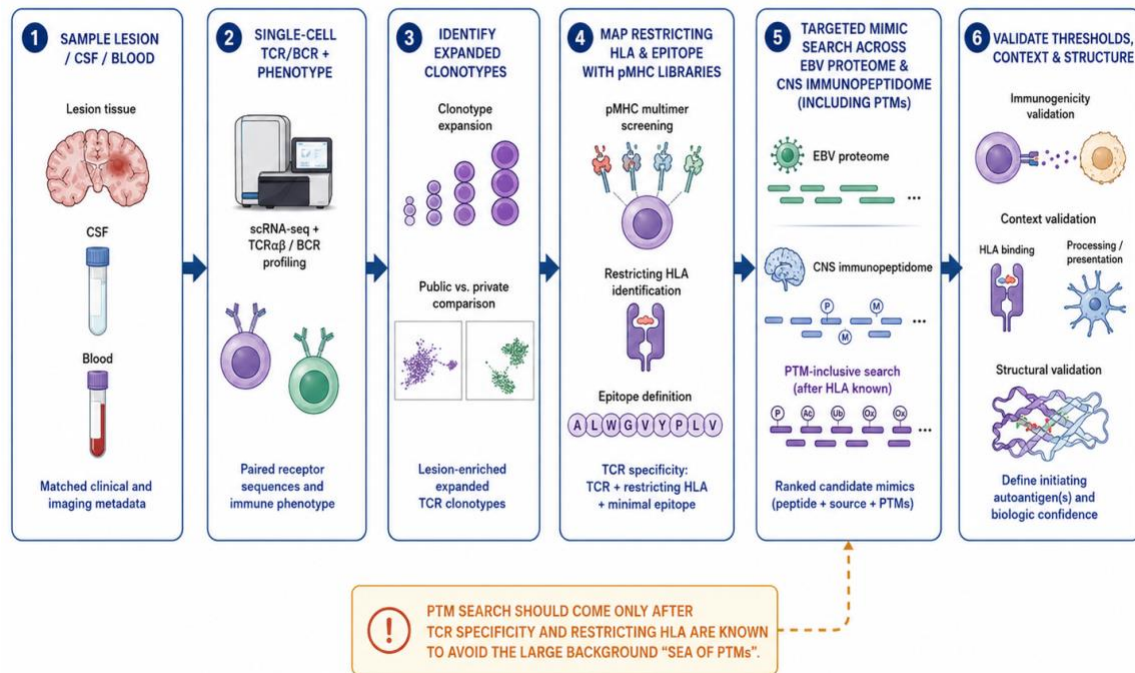


Figure 7. Lesion-anchored, receptor-first workflow for identifying mechanistic MS endotypes. The schematic summarizes six high-level operational phases; the detailed workflow below separates germline, HLA, EBV-strain, and reservoir context as an additional analytic step. Direct viral-self mimicry, EBV-induced altered-self presentation, antiviral surveillance, humoral mechanisms, and niche-dependent B-cell persistence are tested as alternatives or interacting mechanisms. Broad viral, self-proteome, and post-translational-modification searches follow, rather than precede, definition of the receptor and restriction.

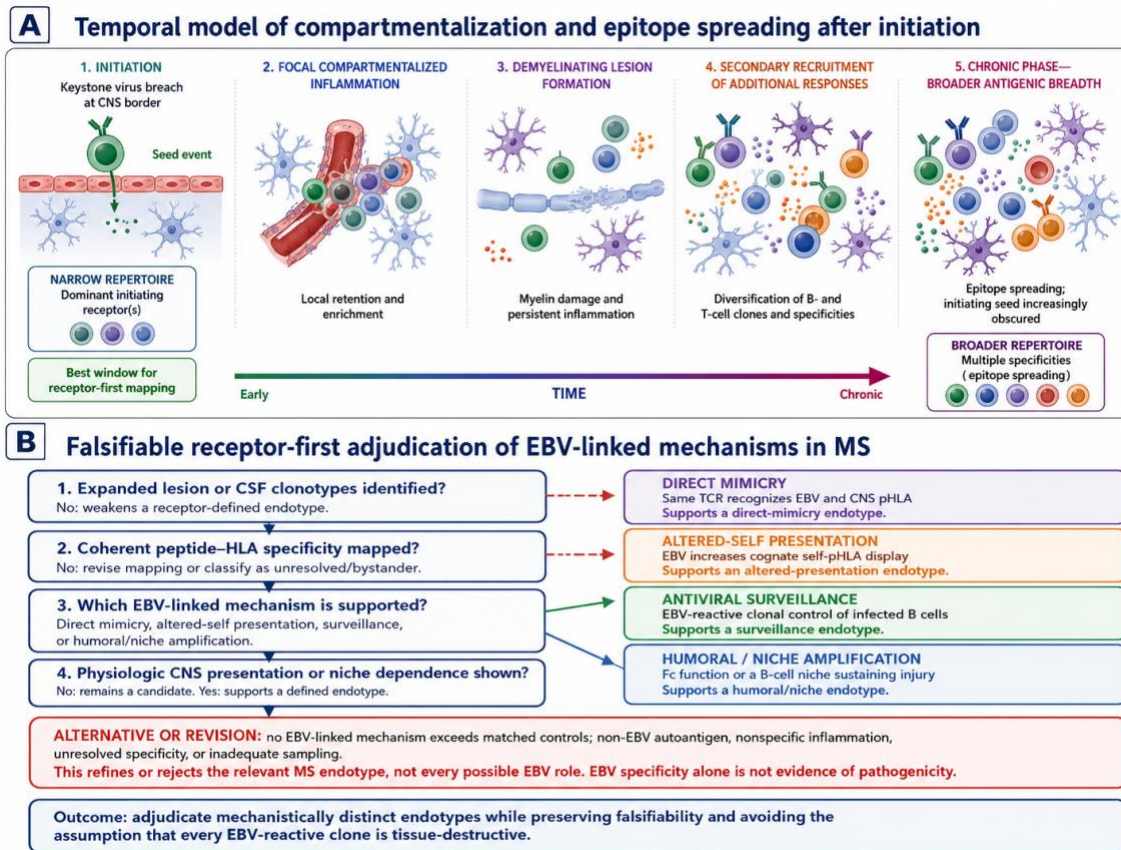


Figure 8. Compartmentalization, epitope spreading, and falsifiable adjudication of EBV-linked mechanisms in MS. Panel A illustrates how secondary recruitment and epitope spreading progressively obscure the initiating receptor and antigenic event, motivating early lesion or CSF sampling. Panel B tests whether expanded clonotypes map to coherent peptide-HLA specificities and whether the evidence supports direct mimicry, altered-self presentation, antiviral surveillance, or humoral/niche amplification at physiologically plausible levels. Negative branches refine or reject the relevant KET-MS endotype rather than treating every EBV-reactive clone as pathogenic.

8.1 Stepwise workflow

1. Define the “time and site of the crime”: obtain active lesion-border tissue when possible, and pair CSF and blood sampling with lesion-stage imaging. Newly forming or gadolinium-enhancing lesions are especially informative for the transient BBB-access phase, whereas chronic lesions are more likely to reflect downstream compartmentalization and epitope spreading^{23,24}. Phenotype infiltrating T cells and B cells using single-cell RNA and paired-chain TCR/BCR analyses.
2. Prioritize expanded clonotypes: identify clonally expanded TCRs (CD4 cytotoxic and CD8) and CSF-resident BCR lineages. Expansion is the strongest prior for relevance.

3. Capture germline, strain, and reservoir context: in parallel with paired-chain TCR recovery, determine host germline TR alleles, high-resolution HLA class I and II genotypes, and, where feasible, EBV strain, EBV DNA or transcript burden, B-cell latency or activation state, and relevant draining or CNS-associated compartments ^{12,22,27,58,66,70}. These variables should be analyzed against distinct endpoints, including pre-onset susceptibility, quantitative reservoir control, and post-onset immune state.
4. Map restricting HLA and epitope: use pMHC-I/II multimer panels, DNA-barcoded peptide–MHC libraries, and functional scanning to identify cognate peptides for dominant TCR clonotypes. Both class I and class II restrictions and latent and lytic EBV antigens should be tested; EBV-reactive clonotypes should then be classified as recognizing infected B cells, CNS targets, or both ^{14,17,21}.
5. Interrogate viral–self relationships only after the receptor and restriction are defined: (i) scan the EBV proteome and autologous EBV sequence for peptides that recreate the relevant pMHC surface; (ii) scan the CNS proteome, including native and PTM peptides, for direct structural mimics; (iii) test whether EBV infection reshapes presentation of the cognate self peptide even in the absence of viral sequence homology; and (iv) confirm presentation by immunopeptidomics under relevant inflammatory and latency conditions ^{18,19}. For humoral pathways, measure both Fab specificity and Fc-receptor-dependent effector function¹³.
6. Validate threshold and access logic: quantify activation thresholds, functional outputs, and the requirement for inflammatory context. Demonstrate that mimic density is sufficient to cross the inhibitory set point in relevant CNS targets or local antigen-presenting cells. In human BBB or neurovascular-unit models, test endothelial activation, adhesion, and transmigration of candidate T-cell clonotypes and B-cell subsets under defined cytokine and antigenic conditions. Where B cells are implicated, test whether antigen capture, T-B collaboration, or LMP1/CD40-like survival signals alter persistence and effector activation within the niche ^{16,82}. Perivascular and meningeal analyses should explicitly test whether local amplification depends on detectable EBV-positive cells or can persist after the initiating viral contribution is no longer locally evident ¹⁵.
7. Close the loop with the original lesion and with structure: re-probe the original lesion or CSF using paired microbial and self or altered-self pHLA reagents; isolate dual-binding cells for paired-chain receptor sequencing, reconstruction, and functional confirmation. Then solve pMHC–TCR (or pMHC–Fab) structures for the keystone and candidate mimic ligands to confirm geometric convergence and avoid sequence-level overinterpretation.

Methodological note: Genome-to-genome and receptor-centric analyses testing HLA, germline TCR, and EBV strain effects must explicitly control for host and viral population structure and shared ancestry. Failure to do so can confound founder effects with true selection or coevolutionary signals and generate spurious associations.

8.2 What would count as a decisive result?

- For a direct-mimicry endotype, a recurrent HLA-stratified viral peptide–HLA surface recognized by expanded lesion or CSF clonotypes, together with demonstrable cross-recognition of a CNS-presented mimic.
- Functional distinction between EBV-reactive clonotypes that mediate antiviral surveillance and those that cross-recognize CNS targets; EBV specificity alone would not establish pathogenicity²¹. A protective HLA-A*02:01-like signal should not be assumed to operate through the same mechanism as a putative HLA-A*03:01 or HLA-B*07 risk signal.
- Evidence that EBV infection reproducibly increases presentation of the cognate self peptide on the relevant HLA molecule, providing an alternative to direct viral–self mimicry¹⁹. For a niche-persistence endotype, demonstrate that antigen-capturing B-cell survival or cognate T–B cooperation is required for sustained local effector activity^{16,82}. For a host-genetic endotype, show whether the relevant HLA effect operates mainly through qualitative antigen specificity or through quantitative control of the EBV reservoir.
- For a humoral endotype, reproducible association of antigen specificity and Fc effector function with lesion activity, followed by experimental evidence that the Fc pathway contributes to tissue injury¹³.
- Demonstration that the CNS mimic is presented in situ (immuno-peptidomics or validated surrogate), and that inflammatory context materially lowers the activation threshold.
- Evidence that EBV strain variation, host modifiers such as vitamin D or smoking, or class I versus class II HLA background shifts the threshold or mimic density required for activation, rather than merely tracking with downstream inflammation.
- Temporal or experimental evidence defining the access sequence at the BBB: direct demonstration that EBV-affected B cells are early entrants would support one mechanistic route, whereas endothelial or T-cell activation followed by secondary B-cell recruitment would support another. Either result should be distinguished from gadolinium enhancement alone, which reports permeability but not the identity of the initiating cell.

9. Translational implications

- Risk prediction and endotyping: extend HLA-only approaches to incorporate high-resolution class I and II genotype, germline TR context, antigen-resolved cellular responses, EBV strain and burden, and functional EBV antibody features^{12 13,14,17,27,70 22}.
- Prevention: EBV vaccination could, in principle, reduce MS incidence by preventing delayed symptomatic primary infection; KET additionally motivates therapeutic or refocusing approaches in established EBV infection (Section 9.1)²⁰.
- Targeted immunotherapy: EBV-specific adoptive T-cell approaches have been explored in progressive MS and provide a translational testbed for EBV-targeting logic¹⁰¹.
- B-cell targeting: the robust clinical effect of B-cell depletion supports a central role for B cells in MS immunopathology^{102,103}. KET interprets this as reducing antigen presentation and keystone-mimic density in the relevant niche. The newer data suggest that relevant B-cell functions may include local T-cell restimulation, HLA-DR15 self-peptide presentation,

antigen capture, and survival within CNS niches ^{16,19,82,83}. Perivascular-cuff data also identify BTK-positive B cells and macrophages as potential compartmental targets, although local EBV positivity was sparse ¹⁵.

- Functional stratification of EBV-reactive lymphocytes: antigen-resolved analysis may distinguish direct cross-reactivity from antiviral surveillance, restricted or exhausted class I responses, late-lytic CD4 responses, and proinflammatory antibody Fc states ^{13,14 17,21}.

9.1 Prophylactic versus therapeutic EBV vaccination in a keystone framework

Recent evidence associating herpes zoster vaccination with lower dementia incidence provides a systems-level precedent for considering neurologic consequences of herpesvirus-directed vaccination. A Wales-based regression-discontinuity analysis found that vaccine eligibility and the resulting increase in uptake were associated with a lower probability of dementia diagnosis over seven years ¹⁰⁴. Complementary observational and quasi-experimental analyses reported similar associations for recombinant zoster vaccine and across different stages of dementia ¹⁰⁵⁻¹⁰⁷. These studies are not randomized experiments in MS pathogenesis and residual confounding remains plausible; their relevance here is that neurologic outcomes may be modified without eradicating a persistent herpesvirus.

KET interprets persistent, co-evolved infections as states in which immune regulation may favor durable containment over sterilization ⁷. Latent herpesvirus infection has conferred heterologous protection in an animal model ¹⁰⁸, and chronic viral infection is increasingly understood as a spectrum of regulated host–pathogen states rather than simply failed clearance ¹⁰⁹. These observations provide a rationale for evaluating both the benefits and unintended consequences of sterilizing vaccine goals in a proposed keystone system.

In the context of EBV and MS, KET raises uncertainty about the optimal immunological endpoint of vaccination. A sterilizing EBV vaccine would remove a near-universal, developmentally timed exposure that may contribute to immune calibration. The population-level consequences of eliminating this interaction are unknown. This perspective does not argue against prophylactic vaccination, particularly given the burden of infectious mononucleosis and EBV-associated malignancy, but cautions against assuming that prevention of infection is necessarily the only desirable endpoint in a co-adapted host–pathogen system.

KET therefore provides a rationale for prioritizing evaluation of therapeutic or response-modifying EBV vaccines alongside prophylactic approaches. Such vaccines could aim to reduce the EBV antigenic set point and follicular amplification while shifting immunodominance away from cross-reactive or otherwise unfavorable responses. A recombinant gp350 vaccine reduced infectious mononucleosis without preventing asymptomatic EBV infection ¹¹⁰, demonstrating that vaccination can modify clinical phenotype without sterilization. Contemporary vaccine development also recognizes the need for multi-antigen and cellular responses rather than neutralizing antibody alone ^{111,112}.

For MS, therapeutic vaccination can be evaluated using mechanistic endpoints over shorter intervals than incidence prevention. Candidate endpoints include intrathecal B-cell and follicular biology; EBV latency and lytic transcription; salivary or compartment-specific shedding; expansion, contraction, and function of antigen-resolved CD4 and CD8 clonotypes; and the specificity and Fc properties of EBV-directed antibodies. Endpoint separation matters here as well: some interventions may improve quantitative control of the EBV reservoir without altering mimic-prone class II programs, whereas others may reshape immunodominance without materially changing viral load. Recent ANO2 and HLA-DR15 data support avoiding amplification of cross-reactive EBNA1-directed clonotypes and reducing EBV-induced self-peptide presentation^{18,19}. Enriched late-lytic CD4 responses and proinflammatory Fc signatures further indicate that vaccine assessment should include cellular antigen breadth, viral shedding, and antibody Fc function rather than titer alone^{13,14}.

Prophylactic EBV vaccination remains important because EBV contributes to infectious mononucleosis, malignancy, and possibly MS. The narrower KET-derived proposal is that therapeutic refocusing, and equilibrium-management strategies deserve prioritized experimental evaluation while prophylactic strategies continue to be developed. Comparative trials should test whether each approach improves viral control, shifts immunodominance away from mimic-prone responses, and benefits MS-relevant immune endotypes without amplifying cross-reactive or proinflammatory programs.

9.2 Diagnostic and endotyping implications

KET suggests that at least a subset of MS may be endotyped by the initiating receptor–pMHC program and the anatomical context in which it becomes pathogenic, rather than by late correlates after epitope spreading. Candidate approaches include:

1. Lesion-anchored receptor mapping: sample CSF during early relapse and active lesion borders when available, reconstruct expanded clonotypes, and map their peptide–HLA specificity.
 2. Mechanism-specific antigen discovery: after defining receptor and restriction, test direct viral–self mimicry, EBV-induced altered self-presentation, and niche-dependent B-cell persistence before broad candidate-antigen or PTM searches.
 3. Host-virus endotyping: integrate HLA class I and II, germline TR alleles, EBV strain or burden, infection timing, and antigen-resolved cellular response profiles, while explicitly distinguishing variables that map to susceptibility, reservoir control, or established disease state.
 4. Longitudinal equilibrium readouts: quantify compartment-specific EBV activity, B-cell state, checkpoint tone, and functional EBV antibody features, including Fcγ-receptor binding and effector function.
- EBV serologic differentiation: persistent high antibody responses to selected EBNA1 peptides were substantially more frequent in MS than in MOG-antibody disease or neuromyelitis optica spectrum disorder and may provide an adjunctive biomarker when

interpreted longitudinally¹¹³. Clinical implementation will require independent validation, HLA-aware calibration, and demonstration of incremental value beyond current diagnostic criteria.

- Preclinical autoantibody endotyping: assays directed at the predictive consensus motif and its candidate EBV and self-antigen sources may identify a highly specific minority endotype suitable for longitudinal surveillance and mechanistic studies^{37,38}. A negative result should not be interpreted as excluding the broader KET-MS model.

9.3 Therapeutic implications of Keystone Epitope Theory in multiple sclerosis

KET directs attention to three mechanistically distinct intervention classes, each of which requires endotype-specific validation:

9.3.1 Modulation of intrathecal antigen amplification

- Target B-cell-rich meningeal and perivascular compartments that sustain antigen presentation, local T–B cooperation, and antibody-secreting-cell activity^{15,82,83}.
- Distinguish depletion of B-cell number from interruption of altered self-peptide presentation, antigen capture, LMP1/CD40-like survival, and BTK-dependent compartmental inflammation^{15,16,19}.

9.3.2 Modulation of EBV-trained effector responses

- Evaluate response-modifying EBV vaccines that alter immunodominance while monitoring direct cross-reactivity, late-lytic CD4 responses, antiviral class I responses, antibody Fc function, and whether the intervention preferentially improves reservoir control, susceptibility-related endotypes, or both^{13,14,17,18}.
- Use EBV-specific cellular therapies only after distinguishing protective antiviral surveillance from candidate pathogenic cross-reactive clonotypes^{17,21,101}. Approaches that mimic an HLA-A*02:01-like surveillance state may be preferable to indiscriminate expansion of all EBV-reactive cells.
- Consider antigen-specific tolerance approaches only after a disease-relevant peptide–HLA and receptor axis is demonstrated in the relevant endotype.

9.3.3 Modulation of activation thresholds within CNS-relevant niches

- Test interventions that restore local inhibitory control or reduce inflammatory amplification without generalized checkpoint manipulation or broad immunosuppression. The key therapeutic question is whether the niche can be pushed back below threshold without sacrificing necessary antiviral containment.

10. Broader implications for autoimmune disease

The endpoint-separated distinction among antigen specificity, control of a persistent antigen source, tissue-niche access, and post-onset immune state may extend well beyond MS. A useful first contrast is between class I-predominant, usually seronegative immunopathology and the

larger group of class II-associated diseases in which a CD4–B-cell response and often a preclinical humoral phase precede or accompany overt tissue injury. This is a heuristic spectrum rather than a binary taxonomy: individual diseases can contain both class I and class II effects, and antibodies may be initiating effectors, amplifiers, protective responses, staging biomarkers, or unrelated epiphenomena.

10.1 Class I-predominant, usually seronegative immunopathology

The MHC-I-opathy framework groups disorders with dominant HLA class I associations, interactions with ERAP1 or ERAP2, relative absence of disease-defining autoantibodies, and weak or secondary class II effects. Prototypic examples include axial spondyloarthritis and ankylosing spondylitis associated with HLA-B*27, HLA-B*27-associated acute anterior uveitis and reactive arthritis, Behçet disease associated with HLA-B*51, psoriasis associated with HLA-C*06:02, and birdshot chorioretinopathy associated with HLA-A*29:02. Psoriatic arthritis and inflammatory-bowel-disease-associated spondyloarthritis also fall within the broader spectrum. These conditions are not all purely autoimmune or exclusively class I-mediated, but they provide natural experiments in which altered immunopeptidomes, direct CD8 or unconventional T-cell recognition, NK/KIR interactions, and tissue-resident effector programs may lie relatively close to pathology ¹¹⁴⁻¹¹⁶.

The evidentiary depth differs substantially within this group. Psoriasis provides a useful receptor-level benchmark: a lesion-derived CD8 TCR recognizes the melanocyte antigen ADAMTSL5 in HLA-C*06:02, and ERAP1 controls generation and abundance of the displayed autoantigen ^{117,118}. This example illustrates an important terminology rule for the broader framework: “HLA-associated” should describe disease-level genetic relationships, whereas “HLA-restricted” should be reserved for experimentally demonstrated receptor-level recognition.

Axial spondyloarthritis and HLA-B*27-associated anterior uveitis are less completely resolved but now include public TCRs recognizing HLA-B*27-bound bacterial and self-peptides and evidence that the relevant CD8 populations carry mucosal differentiation signatures ^{119,120}. These observations strengthen a peptide-specific class I model while leaving the sustaining antigen source, natural tissue display, and relative contributions of direct recognition and enthesial or ocular amplification to be established.

HLA class I-restricted drug hypersensitivities provide additional mechanistic comparators rather than spontaneous autoimmune diseases. Abacavir hypersensitivity associated with HLA-B*57:01, carbamazepine-associated Stevens–Johnson syndrome/toxic epidermal necrolysis associated with HLA-B*15:02, and allopurinol severe cutaneous adverse reactions associated with HLA-B*58:01 demonstrate how a narrowly defined peptide–HLA perturbation can recruit cytotoxic memory without a preceding autoantibody phase ¹²¹⁻¹²³. KET predicts that analogous class I-predominant disorders should be investigated first through lesion-derived receptors, exact HLA restriction, antigen processing, and direct functional testing of microbial, drug-modified, or altered-self pHLA cross-recognition.

10.2 Class II-associated autoimmunity and preclinical humoral organization

MS is more representative of the broader group of autoimmune diseases in which HLA class II risk, CD4–B-cell cooperation, and a clinically silent phase of antigen-specific humoral immunity are prominent. Autoantibodies can precede clinical systemic lupus erythematosus, rheumatoid arthritis, Graves disease, and Hashimoto thyroiditis by years; multiple islet autoantibodies define presymptomatic stages of type 1 diabetes; and tissue-transglutaminase antibodies define celiac autoimmunity in genetically susceptible children before or around clinical diagnosis ¹²⁴⁻¹²⁸. The functional meaning differs by disease. Autoantibodies and immune complexes can directly mediate injury in lupus; autoantibodies may amplify inflammation in rheumatoid arthritis; islet autoantibodies primarily stage a process in which T cells execute beta-cell loss; and celiac tissue damage requires gluten-specific class II-restricted CD4 responses together with intraepithelial effector mechanisms.

For cross-disease comparison, “antibody-executed” should be assigned only when antigen-specific antibody has been shown to produce target dysfunction or injury through passive transfer, receptor modulation, adhesion disruption, complement or Fc dependence, or an equivalent causal experiment. Myasthenia gravis, AQP4-IgG-positive neuromyelitis optica spectrum disorder, and pemphigus provide such mechanistic anchors ¹²⁹⁻¹³¹. This operational definition prevents disease-associated or predictive serology from being mistaken for proof that antibody is the proximal executor of tissue injury.

Celiac disease provides a complementary class II-restricted mimicry example. HLA-DQ2.5-restricted TCRs from patients can recognize both gliadin epitopes and naturally processed bacterial peptides, with structural evidence for a shared recognition solution ¹³². This demonstrates how microbial mimicry can be established at receptor and pHLA level, while also showing that a receptor-defined mimicry mechanism does not by itself establish a persistent keystone reservoir as the sustaining disease antigen.

Class II association does not invariably imply an autoantibody-defined prodrome, and antibody detection does not establish that antibody is the dominant tissue-damaging effector. Narcolepsy type 1, for example, has an exceptionally strong HLA-DQB1*06:02 association without a reproducible disease-defining autoantibody phase ¹³³. In MS, intrathecal oligoclonal IgG, anti-EBNA1 responses, and the predictive autoantibody motif indicate organized humoral immunity, but they do not by themselves identify the initiating self-antigen or prove that antibody is the principal executor of demyelination ^{37,38}. HLA-DRB1*15:01 may organize upstream EBV-directed CD4 and B-cell specificity, direct EBNA1–self cross-reactivity, and EBV-induced presentation of myelin peptides, while class I alleles within the same individual may increase or reduce liability through direct class I-restricted cross-recognition, control of latently infected B cells, or both.

10.3 A general two-HLA-axis, endpoint-separated KET framework

The general framework separates five components that are often collapsed into one disease association: (1) class II-dependent predisposition and CD4–B-cell immune focusing; (2) class I-dependent control of the persistent antigen-source cell, direct cytotoxic tissue recognition, or

both; (3) abundance, state, and anatomical distribution of the microbial or altered-self antigen source; (4) the tissue niche and inhibitory threshold at which recall becomes pathogenic; and (5) post-onset readouts altered by tissue injury, treatment, and epitope spreading. These components need not occur in a rigid sequence. The same HLA allele can affect more than one component, and class I and class II alleles within one patient can have opposing effects.

A complementary decomposition assigns HLA effects to three functions: specificity and focusing, control of the antigen-bearing source or reservoir, and tissue execution. Disease evolution can likewise be separated into imprinting and licensing, later tissue recall, and local amplification. Licensing denotes acquisition of durable effector competence and the regulatory set points governing later recall; it does not imply that the same inflammatory or danger signal must recur when tissue injury begins. Receptor–pHLA recognition selects the antigenic target, whereas cytokine, innate, stromal, endothelial, barrier, and mechanical programs alter the gain, persistence, and anatomical expression of the response.

The four-condition gate should therefore be treated as conjunctive rather than compensatory. Extreme antigen pressure or inflammation cannot substitute for absence of a compatible receptor–HLA context, a relevant ligand at sufficient density, natural presentation in the appropriate tissue niche, or permissive local regulation. This distinction is particularly useful when class I and class II alleles within one individual have opposing effects on susceptibility, reservoir control, and tissue execution.

The general HLA–receptor–ligand–niche architecture should also be distinguished from the stricter KET source hypothesis. A lesion-derived receptor that maps coherently to native or modified self without evidence of persistent-organism imprinting would support antigen-specific disease while weakening the strict keystone explanation for that endotype. Such a result should lead to mechanistic reclassification rather than an indefinitely expanding microbial search.

MS provides an unusually informative test case because a persistent keystone organism is strongly linked to disease susceptibility; HLA-DR15-restricted direct mimicry and altered self-presentation are experimentally accessible; class I responses can be protective surveillance, candidate direct effectors, or markers of antigen pressure; and the CNS provides a defined compartment in which B-cell and myeloid presentation can sustain recall. Comparable studies in rheumatoid arthritis, lupus, type 1 diabetes, autoimmune thyroid disease, celiac disease, narcolepsy, and other class II-associated disorders should test the same components separately rather than assuming that a positive antibody, HLA association, or microbial signal identifies the causal tissue-damaging mechanism.

10.4 General investigative priorities

- Use longitudinal preclinical samples to distinguish initiating immune specificity from responses secondary to tissue injury, treatment, or epitope spreading.
- Resolve exact HLA class I and II alleles, paired TCR or BCR sequences, germline receptor variation, and microbial strain or antigen source within the same individual.

- Test direct microbial–self cross-reactivity separately from pathogen-induced altered self-presentation and from impaired control of the infected or antigen-bearing cell reservoir.
- Determine whether antibodies are causal effectors, inflammatory amplifiers, protective responses, or staging biomarkers, rather than treating seropositivity as a single mechanistic endpoint.
- Demonstrate peptide processing and presentation in the relevant tissue target or local antigen-presenting cell and establish physiologically plausible receptor affinity and antigen density rather than relying on sequence similarity alone.
- Prespecify competing models and negative predictions so that failure of direct mimicry can redirect attention toward altered presentation, reservoir control, humoral effector mechanisms, or non-KET pathways.
- Use “HLA-associated” for disease-level genetic relationships and reserve “HLA-restricted” for receptor-level functional proof in the exact allele context.
- In individuals carrying more than one relevant HLA allele, test within-person allele partition: determine whether one allele primarily shapes antigen-source control while another organizes specificity or tissue execution.
- Interpret negative experiments in relation to tissue access and disease stage. Failure to detect a circuit in blood or in a chronic treated compartment is weaker evidence than an adequately powered negative result in an early active lesion or matched CSF compartment.

These general implications are developed further in a companion review that uses drug hypersensitivity and MS as complementary empirical anchors to separate HLA specificity and focusing, antigen-source control, and tissue execution from the temporal stages of imprinting and licensing, tissue recall, and local amplification ¹⁰. That follow-on analysis also applies the architecture across selected class I-associated, class II-associated, and antibody-executed endotypes and specifies observations that would disconfirm or reassign each proposed mechanism ¹³⁴.

11. Limitations and falsifiability

The utility of Keystone Epitope Theory depends on its capacity to generate testable predictions and to be revised or rejected in light of new data. Figure 8 B summarizes these falsifiability criteria schematically. Observations that would materially challenge the KET-MS framework include:

- Absence of consistently expanded T-cell clonotypes in MS lesions or cerebrospinal fluid, or identification of expanded clonotypes that do not map to coherent peptide–MHC specificities.
- Inability to identify EBV-related or keystone-like epitopes among expanded cytotoxic T-cell clonotypes despite high-sensitivity mapping approaches.
- Evidence that candidate CNS mimic peptides are not presented within relevant anatomical niches, or that observed cross-recognition occurs at affinities or antigen densities unlikely to be physiologically meaningful.

- Reproducible evidence that EBV-specific clonotypes in lesions are antiviral but not cross-reactive would require those clones to be excluded from the pathogenic KET pathway rather than counted as supportive evidence ²¹. Conversely, if protective class I effects such as HLA-A*02:01 prove to act mainly through antiviral containment, they should not be collapsed into the same mechanistic category as class II mimicry effects.
- If EBV-induced self-peptide presentation or B-cell survival mechanisms explain disease without a recurrent viral–self pMHC relationship, a strictly mimic-centered formulation of KET would require expansion ^{16,19}.
- Evidence that local B-cell and CD4 T-cell inflammation persists despite sparse detectable EBV-positive cells would challenge any formulation requiring EBV to be present at every active lesion, while remaining compatible with remote imprinting followed by downstream amplification ¹⁵.
- Late-lytic or EBV-specific responses must be classified as pathogenic, protective, or markers of viral activity; enrichment alone is not sufficient evidence of causal tissue injury ^{14,17}.
- Reports of MS onset before documented primary EBV infection challenge an absolute requirement that EBV precede every case ³³. Failure to confirm remote infection after comprehensive serologic and molecular testing would favor an EBV-independent minority endotype rather than falsify the model for the majority of MS.
- Broader application to autoimmune disease should fail if class I- and class II-associated conditions cannot be separated into reproducible specificity, antigen-pressure, niche, and post-onset-state components, or if those components do not improve prediction or experimental prioritization beyond conventional disease labels.
- Failure to identify a persistent-organism source for an otherwise coherent self-pHLA–receptor circuit should reclassify that endotype as peptide-specific non-KET rather than trigger an unbounded search for additional microbial candidates. This would weaken the strict KET source hypothesis for that endotype without invalidating the more general requirement to resolve receptor, ligand, tissue presentation, and local regulation.
- Current imaging establishes focal BBB permeability as an early event in many new lesions, but does not prove a generalized barrier defect before MS onset or a fixed sequence of BBB leak followed by entry of EBV-positive memory B cells. KET-MS should remain compatible with T-cell-, innate-, or endothelial-initiated access followed by secondary B-cell amplification ²³⁻²⁵.

12. Conclusion

Multiple sclerosis remains an informative unresolved problem in immunology because a highly reproducible association with EBV coexists with marked partial penetrance. EBV exposure is nearly universal, whereas MS develops in only a small subset of exposed individuals. Explanatory models must therefore distinguish liability to develop MS from quantitative control of the persistent EBV reservoir and from immune phenotypes measured after disease onset, while addressing how infection timing, host receptor genetics, viral state, immune regulation, and tissue context combine. Focal BBB activation should be treated as a separate access variable

within this architecture: it can precede conventional lesion evolution, but its initiating cause and temporal relationship to EBV-affected B-cell trafficking remain unresolved.

Within KET, EBV is considered not only as a possible trigger but as a persistent, co-adapted infection that may organize immunodominance and durable control across B-cell, CD4, CD8, and regulatory compartments. Recent studies support several nonexclusive mechanisms: direct EBNA1–ANO2 T-cell cross-reactivity; EBV-induced presentation of myelin peptides on HLA-DR15; survival and local cooperation of antigen-capturing B cells; altered class I antiviral surveillance; enriched late-lytic CD4 responses; and proinflammatory EBV-antibody Fc states¹²⁻¹⁹. These findings broaden KET beyond a single mimicry pathway and suggest that different mechanisms may dominate in different MS endotypes. They also argue against treating HLA-DRB1*15:01 and HLA-A*02:01 as interchangeable “risk genes”: the former may act mainly through qualitative class II biology, whereas the latter is more consistent with protective antiviral surveillance. The CNS threshold gate therefore includes endothelial activation and leukocyte access as well as receptor specificity, antigen pressure, local presentation, and inhibitory control.

A principal investigational implication is to prioritize receptor specificity and cellular context over isolated lists of candidate autoantigens. A receptor-first strategy identifies expanded lesion or CSF clonotypes, defines their HLA restriction and cognate peptide geometry, and then tests direct mimicry, altered self-presentation, antiviral surveillance, antibody function, and niche dependence. This ordering constrains the search space, separates pre-onset liabilities from post-onset correlates, and provides a path toward experimentally defined MS endotypes relevant to diagnosis and therapy.

MS has also informed the formulation of KET. The EBV–MS relationship illustrates that persistent infections are governed by distributed control across HLA class I and II, germline receptor variation, B-cell state, antiviral effector programs, and tissue-specific regulation. MS therefore bridges class II-associated autoimmunity with humoral organization and class I-predominant disorders in which peptide presentation and cytotoxic tissue recognition are more proximal to disease. Pathology may arise when these components become mismatched in timing, antigenic focus, reservoir control, or anatomical location. This perspective supports testing interventions that restore regulated containment or redirect maladaptive responses, while retaining the possibility that some microbe-directed immunity is protective and that some infection-related signatures observed after disease onset are consequences or modifiers rather than initiating causes.

The value of KET depends on its capacity to generate testable predictions and to be revised or rejected as evidence accumulates. Critical evaluations include whether early lesion or CSF receptor mapping converges on coherent HLA-stratified pMHC specificities; whether candidate cross-reactive, altered-presentation, humoral, or niche mechanisms operate at physiologically plausible levels; and whether targeted modification of those mechanisms changes disease biology. Failure of these tests would require revision of the relevant KET-MS endotype. Success would advance mechanistic understanding of MS and provide broader insight into how

persistent infections shape immunodominance, immune equilibrium, and tissue-specific vulnerability.

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